

Nisrine KHNIFASS

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Final-year engineering student at ENSICAEN, specializing in cybersecurity and electronic payment security. Passionate about application security and the protection of payment systems. Currently on an academic exchange in Japan and seeking a 6-month internship starting in March 2027.

PROJECTS

Magecart Detection Scanner (PCI DSS 4.0) - Personal Project
2026 - ongoing
Python, Playwright, SHA-256 | Application security and e-commerce payment compliance

- **Objective:** detect web skimming (Magecart) on e-commerce payment pages and produce a compliance report for PCI DSS 4.0 requirements (6.4.3 script inventory, 11.6.1 tamper detection).
- **Completed:** study of PCI DSS 4.0 requirements and of the Magecart attack cycle (third-party script compromise, obfuscated code injection, card data exfiltration); mapping of requirements to product features; architecture design and MVP definition (script inventory with SHA-256 fingerprints, comparison between two scans, anti-skimming rules, timestamped report).
- **Next steps:** Playwright scanner prototype, local test bench with a harmless skimmer, then a WooCommerce plugin interface.

Cybersecurity Lab Work - ENSICAEN
2024 - 2026

- **Cryptography:** classical ciphers (Caesar, Vigenère) and cryptanalysis, AES / RSA in Python / C, Diffie-Hellman key exchange, hashing and password storage (salting, bcrypt, cracking with John the Ripper / Hashcat)
- **Networking:** packet analysis (Wireshark), port scanning (Nmap), man-in-the-middle attack (ARP spoofing), interception of unencrypted protocols (HTTP, FTP, Telnet), TLS and certificates (OpenSSL)
- **Payments and smart cards:** APDU commands (ISO 7816), EMV transaction flow analysis, ISO 8583 messages, HSM and key management (PIN, KCV), PCI DSS case study

Project Lead - Interactive Escape Game - H3P Solution
2026

- **Led a team of four** using GitLab to design an interactive escape game mobile application in Flutter, with real-time management of multiple teams, an automatically generated map, a timer and interactive questions.
- **Delivered a functional MVP**, validated and positively received by the client.

EXPERIENCE - RESEARCH

Research Intern - Sungkyunkwan University (SKKU)
Apr. - Aug. 2026
Seoul, South Korea | Python, PyTorch, TensorFlow, computer vision, LLMs

- **Objective:** end-to-end prototype detecting household hazards for children from a surveillance camera feed (five-module pipeline, **~48 frames/second**).
- **Results: 88.7 % accuracy** and **100 % recall on dangerous situations** (335 test videos).

EDUCATION

Shibaura Institute of Technology
Sept. 2026 - Feb. 2027
Tokyo, Japan | Academic exchange semester in computer science

ENSICAEN - National Graduate School of Engineering, Caen
2024 - 2027
Caen, France | Engineering Degree in Software Development, specialization in Cybersecurity

- Coursework: Cybersecurity, Cryptography, Software Security, Security Technologies, Coding Theory, Electronic Payments (ISO 7816, EMV, PCI DSS), Software Engineering, Web & Mobile Development, Data Mining & Machine Learning

EM Normandie (dual degree)
2025 - 2027
Caen, France | Master's Degree in Management

Preparatory Classes, Lycée Carnot - Honors
2022 - 2024
Dijon, France | Preparatory classes for competitive engineering school entrance exams (PCSI - PC)

SKILLS

Cybersecurity & Cryptography: symmetric / asymmetric cryptography (AES, RSA, ECC), hashing (SHA-256), PKI & X.509, TLS 1.3, IPsec / VPN, formal verification (ProVerif)

Payment Security: EMV, PCI DSS (6.4.3 / 11.6.1), HSM & key management, ISO 8583, ISO/IEC 7816 / 14443, fraud prevention

Application & Network Security: secure coding, web application security, network security (Wireshark, Nmap, OpenSSL)

Programming: Python, Java, C, C++, JavaScript, Dart, Bash

Frameworks & Data / AI: React, Node.js, Flutter, PostgreSQL, PyTorch, TensorFlow, machine learning

Tools & Methods: Git, GitLab, GitHub, Linux, design patterns (SOLID), Agile

Languages: French (native), English (fluent, TOEIC 900), Spanish (intermediate), German (beginner), Japanese (beginner)

OTHER PROFESSIONAL EXPERIENCE

Sales Associate - C&A
Aug. 2024 - July 2025

- Independently managed a retail department: restocking, inventory checks, customer advice and assistance

Property Rental Coordinator - Self-employed
2022 - 2024

- Managed apartment rentals, contract tracking, scheduling and administrative tasks

ADDITIONAL INFORMATION

Interests: Entrepreneurship, Travel, Sports, Cooking | **Other:** Driving license (Category B), own vehicle, BAFA (French youth activity leader certificate)